

NODE.JS + EXPRESS

AWS EC2 Dynamic Website Deployment

Project Overview

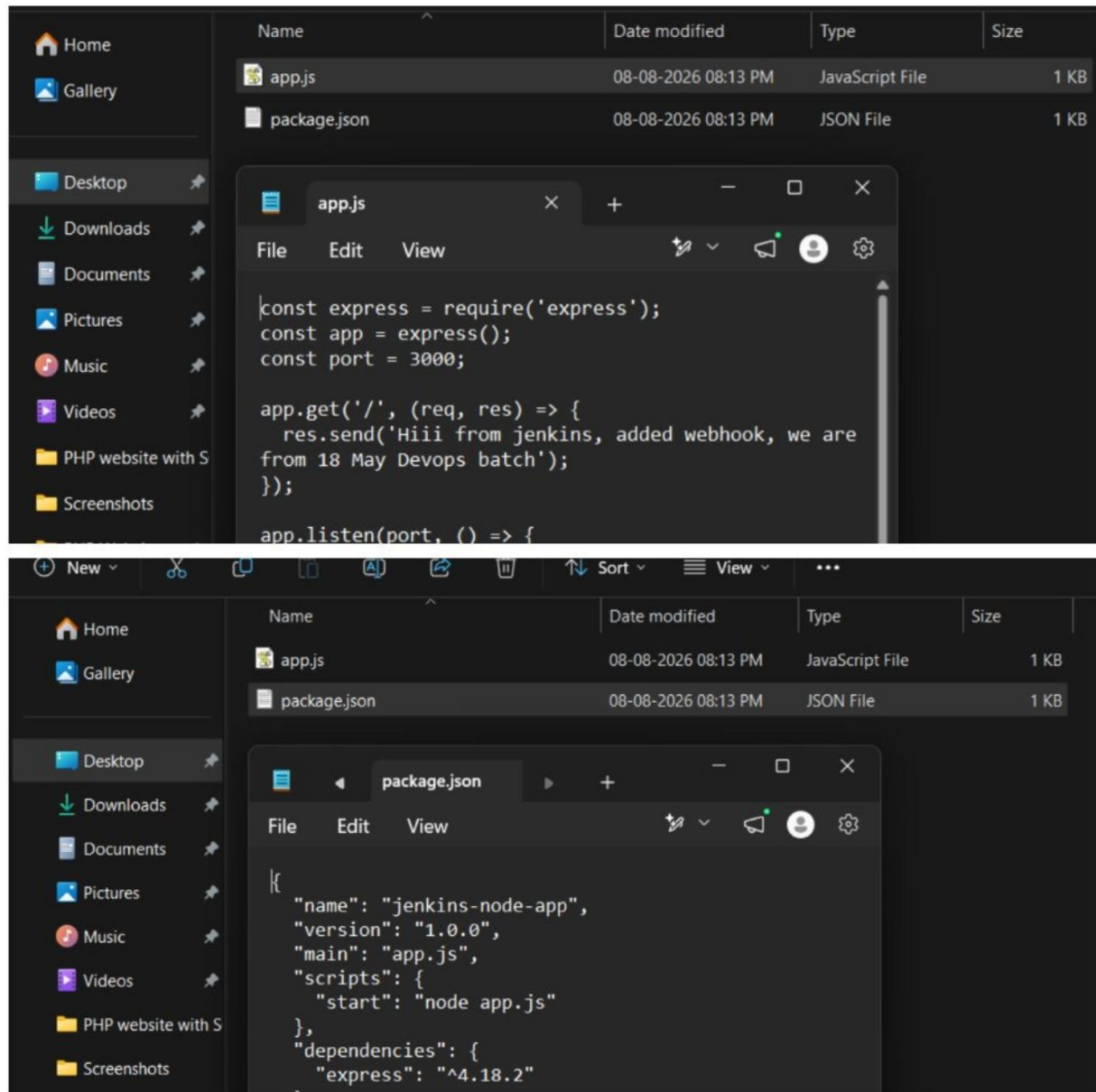
Objective: Deploy a Node.js + Express application from a developer laptop to GitHub and then to AWS EC2. The EC2 server clones the repository, installs dependencies with npm and runs the application on port 3000.

Application Structure

Before npm install	After npm install
<code>app.js</code> <code>package.json</code>	<code>app.js</code> <code>package.json</code> <code>package-lock.json</code> <code>node_modules/</code>

Troubleshooting Note: `sudo git clone` made the repository root-owned. This caused the initial npm EACCES error; ownership was corrected with `sudo chown -R ec2-user:ec2-user .` before running `npm install`.

Source files created on the developer laptop.



Git initialized, committed and pushed to GitHub.

```
MINGW64:/c/Users/praje/OneDrive/Desktop/NODE JS Website Files

praje@Prem_07 MINGW64 ~/OneDrive/Desktop/NODE JS Website Files
$ git init
Initialized empty Git repository in C:/Users/praje/OneDrive/Desktop/NODE JS Website Files/.git/

praje@Prem_07 MINGW64 ~/OneDrive/Desktop/NODE JS Website Files (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    app.js
    package.json

nothing added to commit but untracked files present (use "git add" to track)

praje@Prem_07 MINGW64 ~/OneDrive/Desktop/NODE JS Website Files (master)
$ git add .

praje@Prem_07 MINGW64 ~/OneDrive/Desktop/NODE JS Website Files (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   app.js
    new file:   package.json

praje@Prem_07 MINGW64 ~/OneDrive/Desktop/NODE JS Website Files (master)
$ git commit -m "tested ok"
[master (root-commit) 604b206] tested ok
 2 files changed, 22 insertions(+)
 create mode 100644 app.js
 create mode 100644 package.json

praje@Prem_07 MINGW64 ~/OneDrive/Desktop/NODE JS Website Files (master)
$ git branch main -M

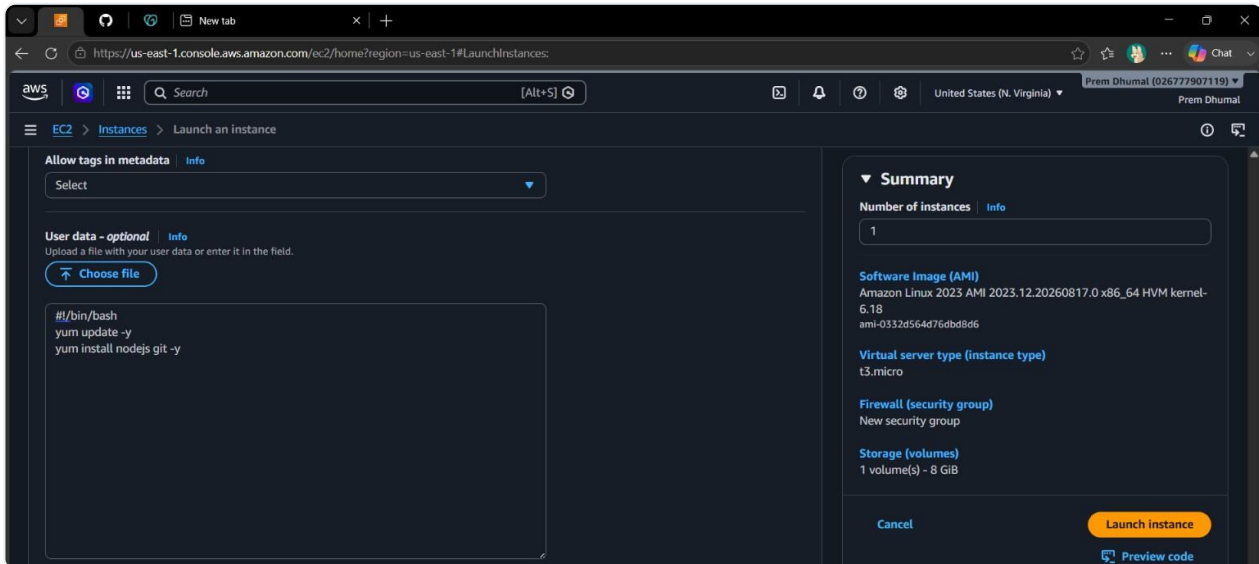
praje@Prem_07 MINGW64 ~/OneDrive/Desktop/NODE JS Website Files (main)
$ git remote add origin https://github.com/premcloud7/NodeJS-Express-Dynamic-Website.git

praje@Prem_07 MINGW64 ~/OneDrive/Desktop/NODE JS Website Files (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 565 bytes | 282.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/premcloud7/NodeJS-Express-Dynamic-Website.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

03

GitHub Repository

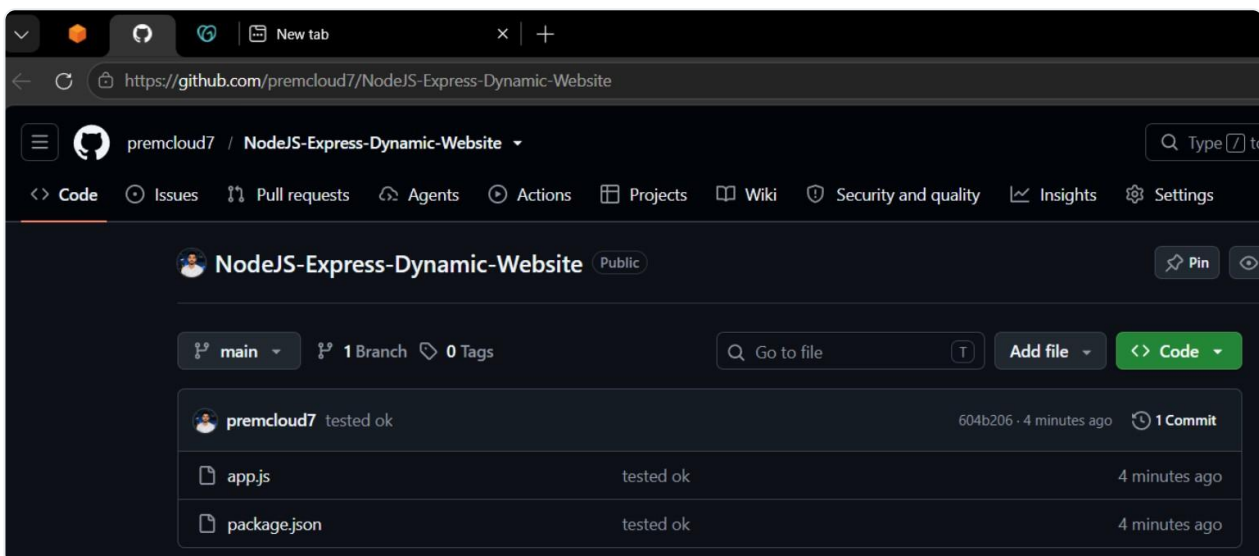
Remote repository containing app.js and package.json.



04

EC2 Instance Configuration

Amazon Linux 2023, t3.micro and EC2 User Data configuration.



05

EC2 Server Status

EC2 instance running with passed status checks.

```

ec2-user@ip-172-31-11-225:~
PS C:\Users\praje> ssh -i '...\Downloads\second server.pem' ec2-user@18.207.111.146
#_
~_#####_ Amazon Linux 2023
~~\#####\
~~\###|
~~\#/
~~V~'---> https://aws.amazon.com/linux/amazon-linux-2023
~~~
~~~
~~~
~~~
Last login: Tue Aug 18 14:19:29 2026 from 152.58.33.19
[ec2-user@ip-172-31-11-225 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-11-225 ~]$ node --version
v18.20.8
[ec2-user@ip-172-31-11-225 ~]$ npm --version
10.8.2
[ec2-user@ip-172-31-11-225 ~]$ git --version
git version 2.50.1

```

06

Server Environment Verification

SSH session with Node.js, npm and Git version verification.

The screenshot shows the AWS Management Console for the EC2 Instances page. The instance 'NodeJS Express Server' (i-0eee19d5e1f05cbcb) is in a 'Running' state. The console displays details such as the Public IPv4 address (18.207.111.146), Private IPv4 addresses (172.31.11.225), and the Public DNS (ec2-18-207-111-146.compute-1.amazonaws.com). The instance is a t3.micro type in the us-east-1a availability zone.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
NodeJS Express...	i-0eee19d5e1f05cbcb	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-18-207-111-146.co...

I-0eee19d5e1f05cbcb (NodeJS Express Server)

- Instance summary**
 - Instance ID: i-0eee19d5e1f05cbcb
 - IPv6 address: -
 - Public IPv4 address: 18.207.111.146 | open address
 - Private IPv4 addresses: 172.31.11.225
 - Instance state: Running
 - Public DNS: ec2-18-207-111-146.compute-1.amazonaws.com | open address

07

Application Repository Cloning

Repository cloned from GitHub and files verified on EC2.

```
ec2-user@ip-172-31-11-225:~/ × + ▼
[ec2-user@ip-172-31-11-225 ~]$ ls
[ec2-user@ip-172-31-11-225 ~]$ sudo git clone https://github.com/premcloud7/NodeJS-Express-Dynamic-Website.git
Cloning into 'NodeJS-Express-Dynamic-Website'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 4 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
[ec2-user@ip-172-31-11-225 ~]$ ls
NodeJS-Express-Dynamic-Website
[ec2-user@ip-172-31-11-225 ~]$ cd NodeJS-Express-Dynamic-Website/
[ec2-user@ip-172-31-11-225 NodeJS-Express-Dynamic-Website]$ ls
app.js  package.json
[ec2-user@ip-172-31-11-225 NodeJS-Express-Dynamic-Website]$ |
```

08

Application Dependency Error

Expected missing Express dependency error before npm install.

```
ec2-user@ip-172-31-11-22: × + ▼
[ec2-user@ip-172-31-11-225 NodeJS-Express-Dynamic-Website]$ node app.js
node:internal/modules/cjs/loader:1143
  throw err;
  ^

Error: Cannot find module 'express'
Require stack:
- /home/ec2-user/NodeJS-Express-Dynamic-Website/app.js
   at Module._resolveFilename (node:internal/modules/cjs/loader:1140:15)
   at Module._load (node:internal/modules/cjs/loader:981:27)
   at Module.require (node:internal/modules/cjs/loader:1231:19)
   at require (node:internal/modules/helpers:177:18)
   at Object.<anonymous> (/home/ec2-user/NodeJS-Express-Dynamic-Website/app.js:1:17)
   at Module._compile (node:internal/modules/cjs/loader:1364:14)
   at Module._extensions..js (node:internal/modules/cjs/loader:1422:10)
   at Module.load (node:internal/modules/cjs/loader:1203:32)
   at Module._load (node:internal/modules/cjs/loader:1019:12)
   at Function.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:128:12) {
  code: 'MODULE_NOT_FOUND',
  requireStack: [ '/home/ec2-user/NodeJS-Express-Dynamic-Website/app.js' ]
}

Node.js v18.20.8
```


09

Node.js Dependency Installation

Permission corrected, dependencies installed and application started.

```
ec2-user@ip-172-31-11-225:~/ NodeJS-Express-Dynamic-Website$ sudo chown -R ec2-user:ec2-user .
[ec2-user@ip-172-31-11-225 NodeJS-Express-Dynamic-Website]$ ls -ld
drwxr-xr-x. 3 ec2-user ec2-user 52 Aug 18 14:26 .
[ec2-user@ip-172-31-11-225 NodeJS-Express-Dynamic-Website]$ npm install

added 68 packages, and audited 69 packages in 3s

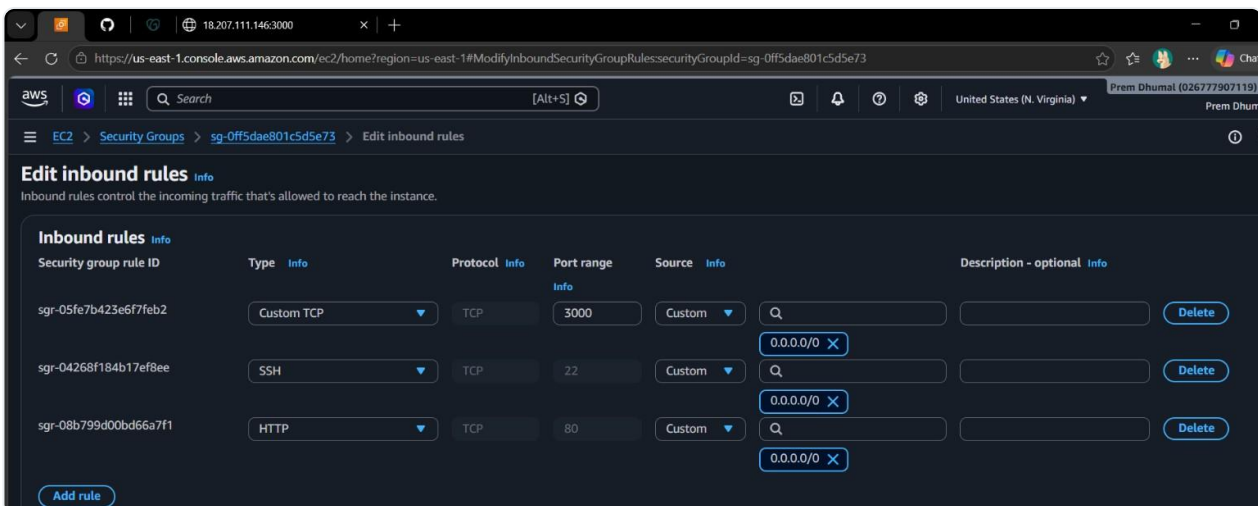
15 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
[ec2-user@ip-172-31-11-225 NodeJS-Express-Dynamic-Website]$ node app.js
App listening at http://localhost:3000
|
```

10

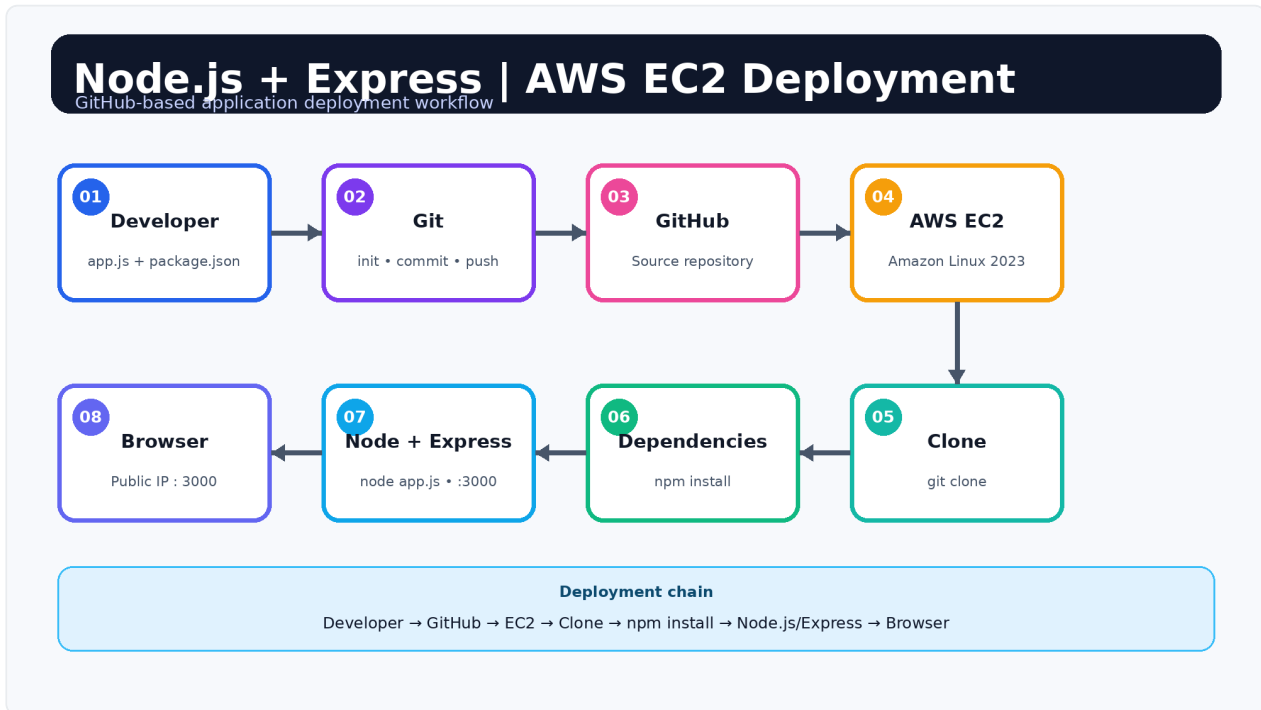
Security Group Configuration

TCP port 3000 allowed for the Node.js application.



Deployed Node.js Application

Final Express application opened through the EC2 public IP.



Deployment Complete

The final browser test confirms the Express application is reachable on port 3000.

Final flow: GitHub → EC2 → npm install → node app.js → :3000 → Browser